

Flight Operations Analytics - Executive Overview

Trusted operating profile, review priorities and decision-support scope

Overview

Distributions

Network & Routes

Fleet & Payload

Predictive Review

Data Quality

15,931
Flight Count

147
Distinct Routes

349
Data Quality Review Flights

2.2%
Data Quality Review Rate

1. Operating Envelope

- **P50** is the value at or below which **50%** of selected flights fall.
- **P85** is the value at or below which **85%** fall.

-3 / 22 kts
Wind P50 / P85

593 / 1,105 NM
Distance P50 / P85

2. Network & Quality

- Review attention spans **88** routes.
- Open the **route queue** to identify where validation is concentrated.

728 NM
Average Distance

88
Routes with Review Records

3. Fleet & Payload

- These records **require** validation. High TOW utilisation and payload-review flags are **not** safety or performance conclusions.

44
Above 95% TOW Utilisation

128
Payload Review Flights

4. Predictive Review

- **Residuals** support **analytical triage** only; the model is **not** operationally certified.

Linear Regression
Selected Model

633.7 | 0.992
Holdout RMSE | R²

Trusted Data Foundation

15,931 reconciled flight records | 0 hard failures | 349 review flights retained

Filters

Clear all filters

Date Range

6/1/202412/31/2024

Airline

All

Aircraft Model

All

Origin

All

Destination

All

Route

All

Flight Operations Analytics - Wind & Distance Distributions

Wind and distance distributions, cumulative coverage and percentile benchmarks

Overview

Distributions

Network & Routes

Fleet & Payload

Predictive Review

Data Quality

15,931

Flight Count

15,931

Wind Eligible Flights

15,931

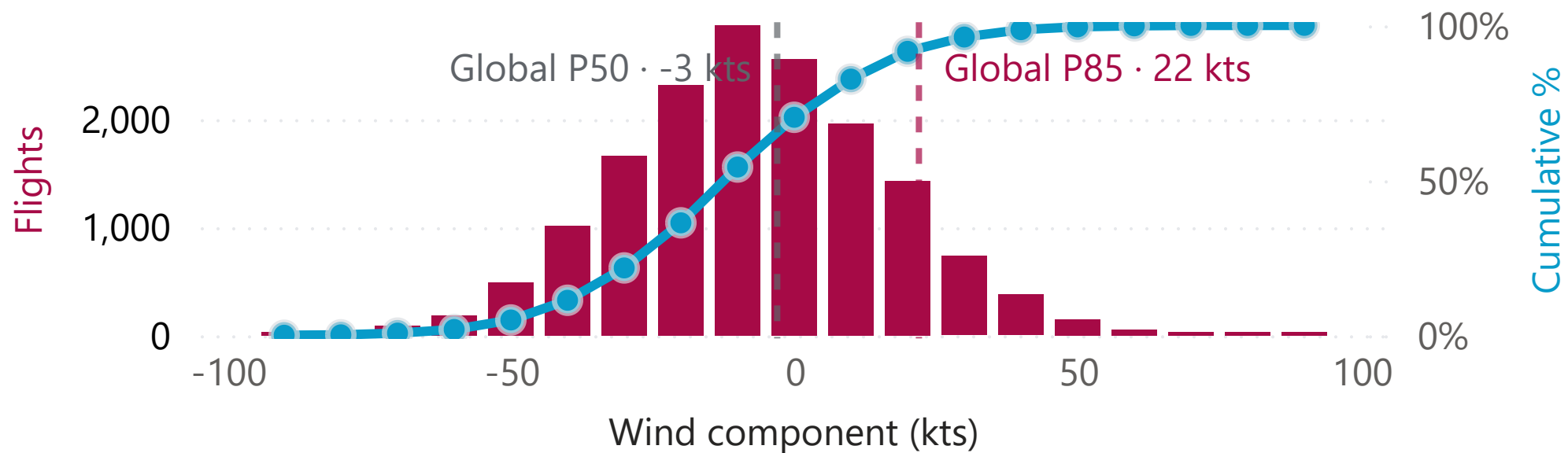
Distance Eligible Flights

2.2%

Data Quality Review Rate

Wind Distribution and Cumulative Coverage

Bars show flights per 10-knot band; the line shows cumulative coverage



-3

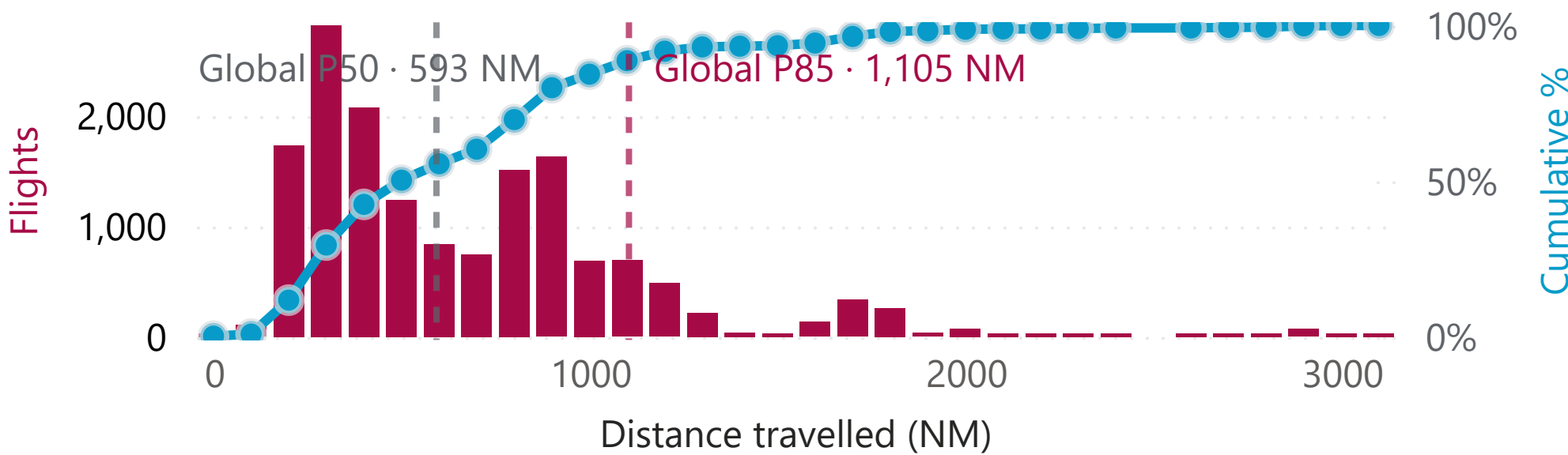
Selected Wind P50 (kts)

22

Selected Wind P85 (kts)

Distance Distribution and Cumulative Coverage

Bars show flights per 100-NM band; the line shows cumulative coverage



593

Selected Distance P50 (NM)

1,105

Selected Distance P85 (NM)

Flights in Selected Distribution Band

Select a wind or distance band to inspect the contributing flight records

Flight Date	Airline	Route	Aircraft Config.	Wind (kts)	Distance (NM)	Planned TOW	Planned Payload	Data Quality Status
02 Jun 2024	EW	AGP-DUS	A321-91000	-16	1,099	81,902	21,682	No Review Warning
02 Jun 2024	EW	AGP-DUS	A321-91000	-14	1,099	81,078	21,323	No Review Warning
02 Jun 2024	EW	ALC-DUS	A320-71500	-15	996	67,560	16,399	No Review Warning
02 Jun 2024	EW	ATH-DUS	A319-69500	-10	1,167	63,010	12,925	No Review Warning

Decision Guide

P50 is the point below which 50% of selected flights fall. **P85** is the point below which 85% fall and supports planning scenarios. Review flights in the highest bands before making conclusions. Wind and distance alone do not prove delay, fuel use, or operational risk.

Filters

Clear all filters

Date Range

6/1/2024



12/31/2024



Airline

All



Aircraft Model

All



Route

All



Page Filters

Origin

All



Destination

All



Data Quality Status

All



In case of questions, please contact:

vikasrathoreofficial@gmail.com

Upstream Input Sources:

Data for Analysis.xlsx, OurAirports airports.csv

Report Data Source (semantic model):

SM_EW_Flight_Operations

Last Refresh:

15 Jul 2026, 14:33 UTC

Flight Operations Analytics - Network & Route Priorities

Route concentration, operating patterns and record-level investigation

Overview

Distributions

Network & Routes

Fleet & Payload

Predictive Review

Data Quality

Filters

Clear all filters

Date Range

6/1/2024



12/31/2024



Airline

All



Aircraft Model

All



Route

All



Page Filters

Origin

All



Destination

All



15,931

Flight Count

147

Distinct Routes

728

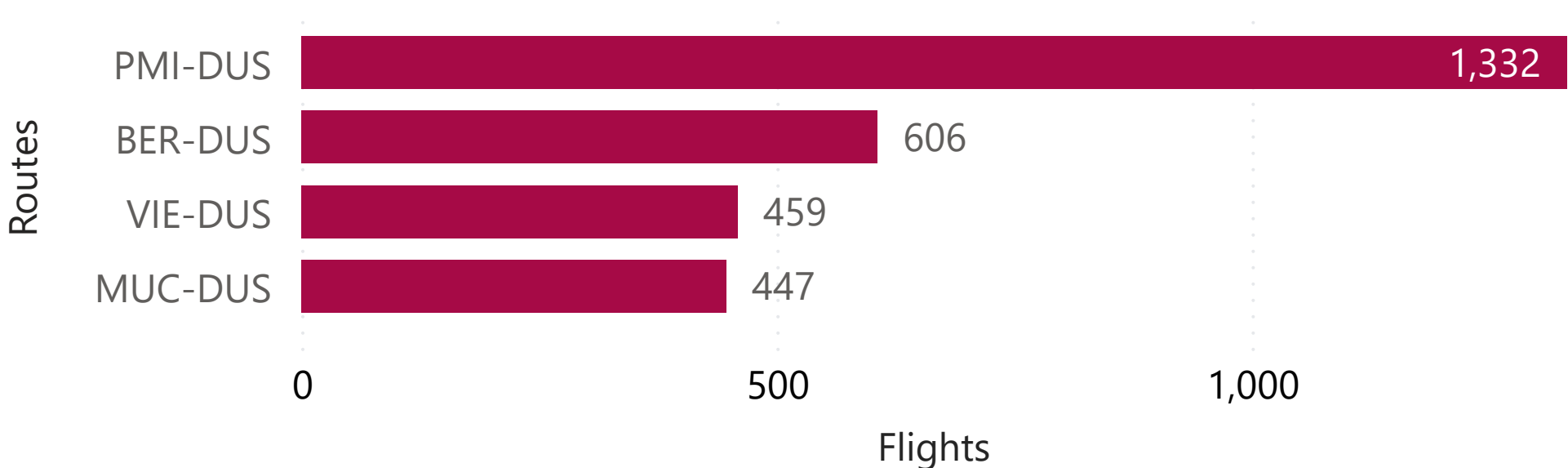
Average Distance (NM)

88

Routes with Data Quality Review

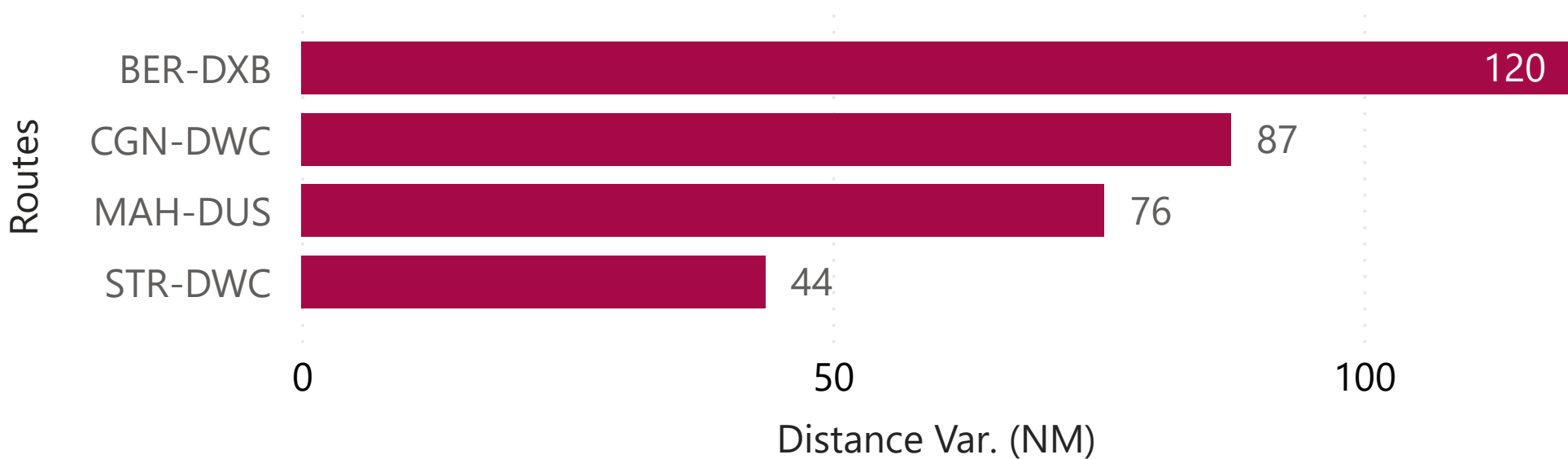
Top Routes by Flight Volume

Top 10 routes by selected flight count



Routes with Highest Distance Variability

Top 10 routes with at least 10 selected flights



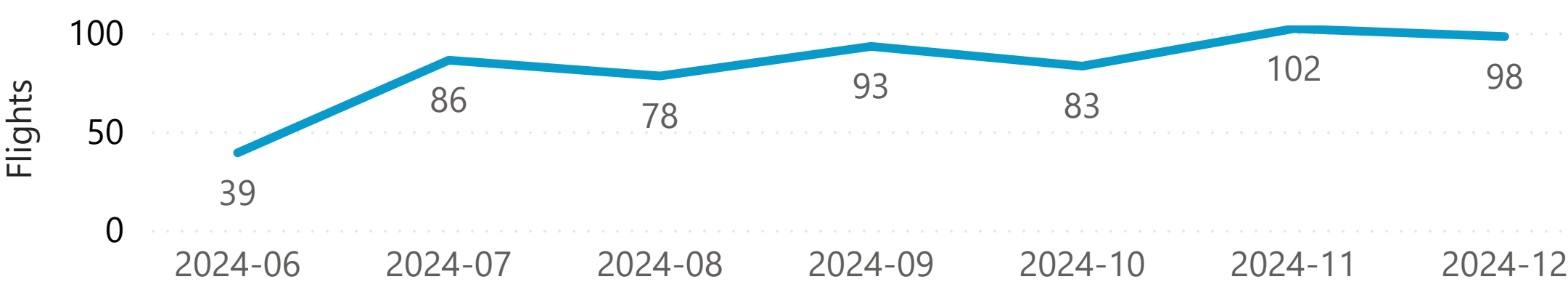
Route Investigation Queue

Compare route vol., dist. variation & data-quality review concentration

Route	Flights	Sample Share	Avg. Distance	Distance Variability	Review Flights	Review Rate	Aircraft Configs.
BER-DUS	606	3.8%	282	10	27	4.5%	6
CGN-DUS	38	0.2%	98	9	21	55.3%	6
STR-DUS	18	0.1%	241	6	15	83.3%	4
VIE-DUS	459	2.9%	480	21	15	3.3%	6
PMI-DUS	1,332	8.4%	882	32	14	1.1%	6
AGB-DUS	250	1.6%	1,100	24	12	4.8%	6
Total	15,824	100.0%	728	458	312	2.0%	6

Observed Monthly Flight Profile

Selected flight volume by month; select a route above to focus the profile



Decision Guide

Prioritize routes combining **high vol.**, **dist. variability** or **review concentration**. Validate low-sample routes before comparison. Sample share is **not** market share or route

Flight Operations Analytics - Fleet, Weight & Payload

Aircraft deployment, planned operating-envelope utilisation and payload relationships

Overview

Distributions

Network & Routes

Fleet & Payload

Predictive Review

Data Quality

80%

Average TOW Utilisation

15,806

Average Remaining MTOW Margin

12,061

Average Planned Payload

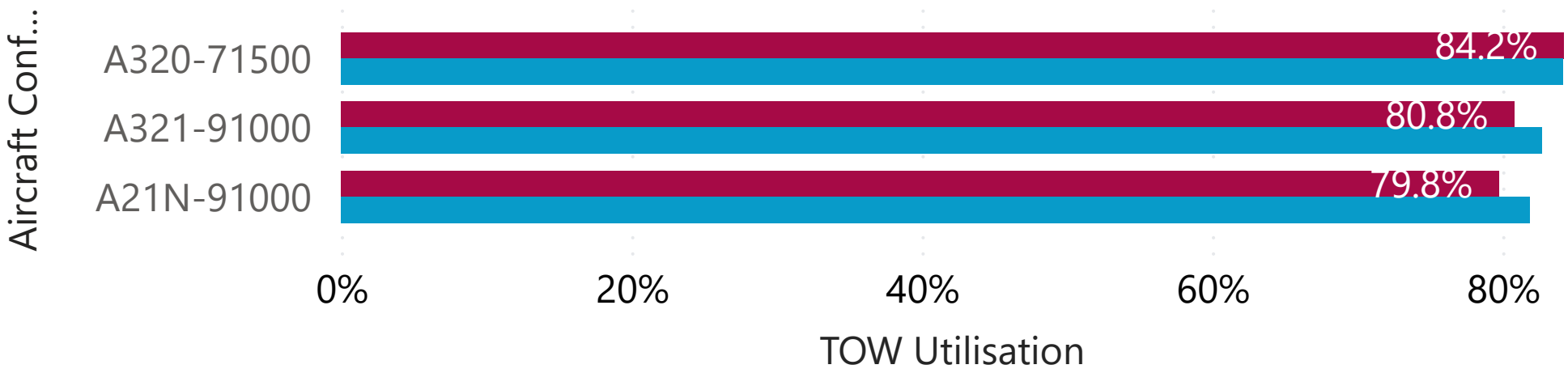
44

High TOW Utilisation Flights

TOW Utilisation by Aircraft Configuration

Average and median supplied planned-TOW utilisation

● Average TOW Utilisation ● Median TOW Utilisation



High TOW Utilisation Review

44 selected flights above the governed 95% utilisation threshold

Flight Date	Route	Aircraft Configs.	Planned TOW	Aircraft MTOW	TOW Utilisation	Remaining MTOW
26 Oct 2024	LCA-DUS	A320-71500	70,023	71,500	97.9%	
25 Jul 2024	ATH-DUS	A320-71500	69,035	71,500	96.6%	
23 Nov 2024	LCA-DUS	A320-71500	69,023	71,500	96.5%	
13 Oct 2024	BCN-DUS	A319-69500	67,026	69,500	96.4%	
09 Nov 2024	AGP-DUS	A320-71500	68,939	71,500	96.4%	
Total			87,119	91,000	97.9%	

Planned Payload vs Passenger Count

Representative pattern across payload-analysis-eligible flights

● Airbus A319ceo ● Airbus A320ceo ● Airbus A320neo ● Airbus A321ceo ● Airbus A321...



Aircraft Deployment by Route

Observed configuration mix across the top 10 selected routes

Route	A20N-81000	A21N-91000	A319-69500	A320-71500	A320-81000	A321-91000
PMI-DUS	82	88	169	139	588	266
BER-DUS	28	14	262	46	233	23
VIF-DUS	12	7	163	56	205	16
Total	293	169	1,577	521	2,207	386

Decision Guide

Prioritise **high-utilisation flights** and **concentrated route-configuration patterns** for validation. These signals support review, not safety or performance conclusions.

Filters

Clear all filters

Date Range

6/1/2024



12/31/2024



Airline

All



Aircraft Model

All



Route

All



Page Filters

Aircraft Configuration

All



TOW Utilisation Status

All



Payload Review Status

All



Flight Operations Analytics - Predictive Analytics

Planned TOW model performance, explainability and residual review

Overview

Distributions

Network & Routes

Fleet & Payload

Predictive Review

Data Quality

1819

Model Scored Flights

398.2

Model MAE

634

Model RMSE

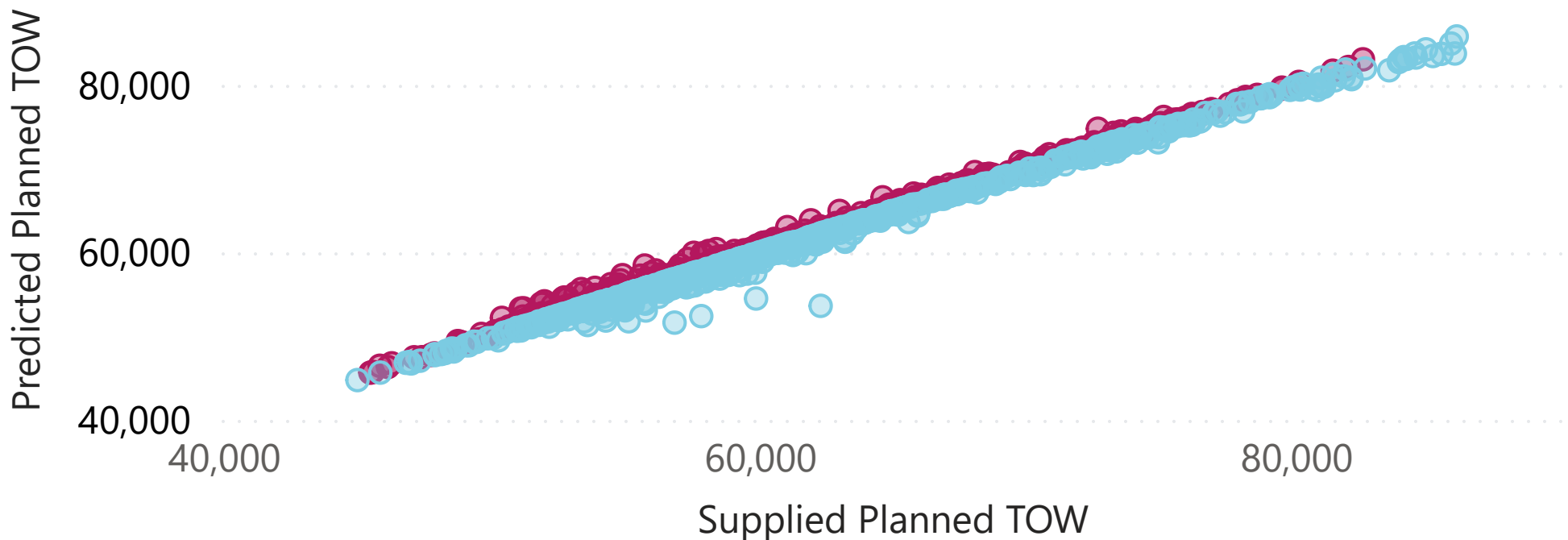
0.992

Model R²

Supplied vs Predicted Planned TOW

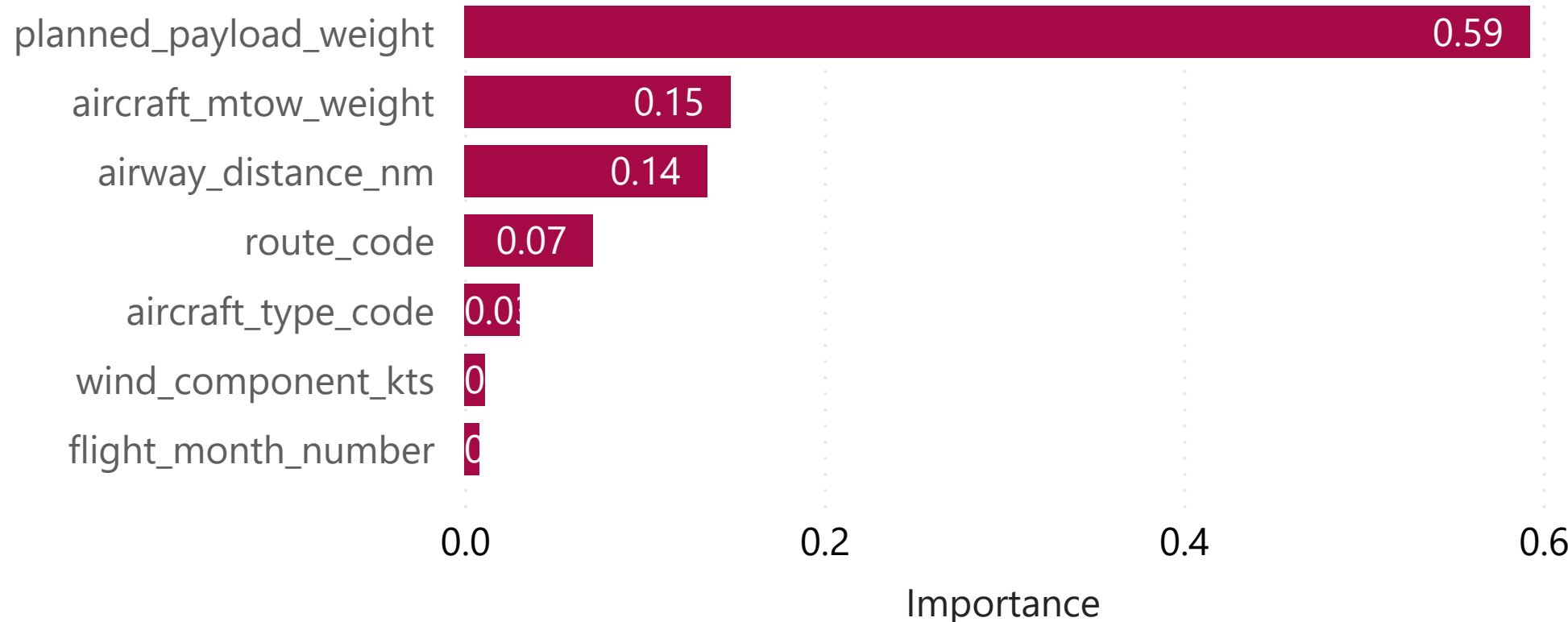
Each point represents one scored flight

● OVER_PREDICTED ● UNDER_PREDICTED



GBT Logical Feature Importance

Normalized model association; importance does not establish causality



Flights Ranked by Absolute Residual

Largest differences between supplied and predicted planned TOW appear first

Flight Date	Airline	Route	Aircraft Configs.	Supplied Planned TOW	Predicted Planned TOW	Absolute Residual	Prediction Direction
20 Dec 2024	Eurowings Europe	HAM-DUS	A320-81000	62,289.00	53,613.74	8,675.26	UNDER_PREDICTED
16 Dec 2024	Eurowings Europe	MUC-DUS	A319-69500	57,835.00	52,385.32	5,449.68	UNDER_PREDICTED
17 Dec 2024	Eurowings	MUC-DUS	A319-69500	59,880.00	54,484.62	5,395.38	UNDER_PREDICTED
18 Dec 2024	Eurowings	HAM-DUS	A319-69500	56,839.00	51,608.35	5,230.65	UNDER_PREDICTED

Decision Guide

Use **absolute residuals** to **prioritise flights** for investigation. **Validate** data quality and aircraft/route context first. **GBT feature** importance shows model association, **not** causality. This prototype **does not support** dispatch, safety or certification decisions.

Filters

Clear all filters

Date Range

6/1/2024



12/31/2024



Airline

All



Aircraft Model

All



Route

All



Page Filters

Model Name

Linear Regression



Prediction Direction

All



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Flight Operations Analytics - Data Quality & Trust

Rule-based review, affected records, analysis eligibility and trusted lineage

Overview

Distributions

Network & Routes

Fleet & Payload

Predictive Review

Data Quality

15,931

Trusted Flights

349

Review Flights

358

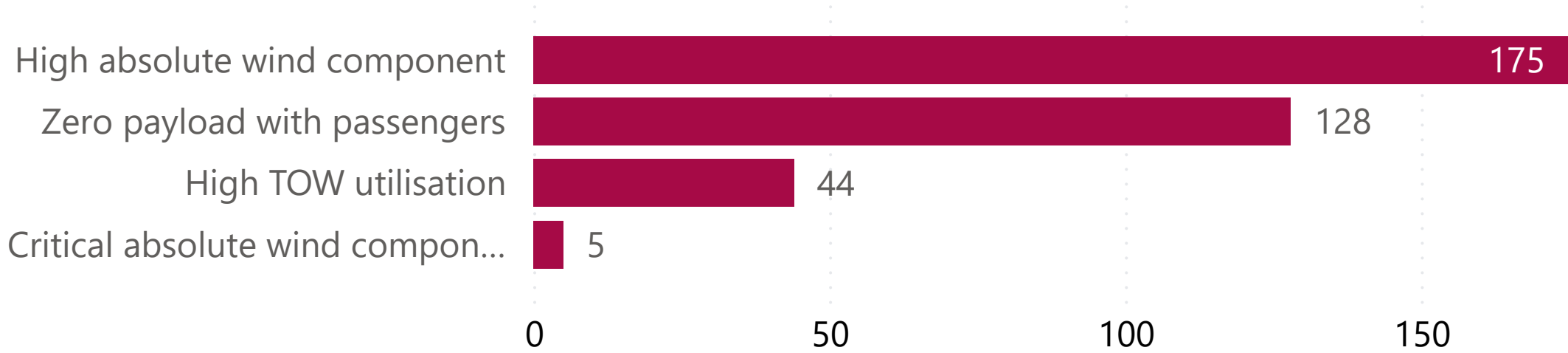
Rule Triggers

0

Hard Failures

Data-Quality Rule Triggers

Select a rule to view its definition and the exact affected flight records



Global Analysis Exclusions by Use Case

Shows global analysis exclusions from data-quality rules, independen...



Select a rule to view its definition and exact affected records.

Records Behind the Selected Data-Quality Finding

Select a rule above to inspect the exact trigger records and their analytical impact

Flight Date	Airline	Route	Aircraft Config.	Review Rule	Rule Code	Severity	Observed Value	Expected Condition
10 Oct 2024	EW	STR-DUS	A320-81000	Zero payload with passengers	DQ101	Warning	Payload=0 PAX=96	planned_payload_weig
11 Oct 2024	4U	STR-DUS	A319-69500	Legacy airline code	DQ106	Warning	4U Germanwings	carrier_status = CURRE
11 Oct 2024	EW	PRN-DUS	A320-71500	High TOW utilisation	DQ105	Warning	0.9542797202797203	tow_utilisation_pct <=
12 Oct 2024	EW	STR-DUS	A319-69500	Zero payload with passengers	DQ101	Warning	Payload=0 PAX=98	planned_payload_weig
12 Oct 2024	EW	PMO-	A320-71500	High TOW utilisation	DQ105	Warning	0.950979020979021	tow_utilisation_pct <=

Decision Guide

Review warnings are **not** automatic deletions. Exclude a flight only from analyses affected by the rule.

Hard failures are the **only** Trusted exclusion criterion. Check the record and source context before correction or escalation.

Filters

Clear all filters

Date Range

6/1/2024



12/31/2024



Airline

All



Aircraft Model

All



Route

All



Page Filters

Quality Dimension

All



Affected Analysis

All



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Upstream Input Sources:

Data for Analysis.xlsx, OurAirports airports.csv

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SM_EW_Flight_Operations

Last Refresh:

15 Jul 2026, 14:33 UTC